

CHAPTER 13

SEISMIC DESIGN REQUIREMENTS FOR NONSTRUCTURAL COMPONENTS

13.1 GENERAL

13.1.1 Scope. This chapter establishes minimum design criteria for nonstructural components that are permanently attached to structures and for their supports and attachments. Where the weight of a nonstructural component is greater than or equal to 25% of the effective seismic weight, W , of the structure as defined in Section 12.7.2, the component shall be classified as a nonbuilding structure and shall be designed in accordance with Section 15.3.2.

13.1.2 Seismic Design Category. For the purposes of this chapter, nonstructural components shall be assigned to the same Seismic Design Category as the structure that they occupy or to which they are attached.

13.1.3 Component Importance Factor. All components shall be assigned a component Importance Factor as indicated in this section. The component Importance Factor, I_p , shall be taken as 1.5 if any of the following conditions apply:

1. The component is required to function for life-safety purposes after an earthquake, including fire protection sprinkler systems and egress stairways.
2. The component conveys, supports, or otherwise contains toxic, highly toxic, or explosive substances where the quantity of the material exceeds a threshold quantity established by the Authority Having Jurisdiction and is sufficient to pose a threat to the public if released.
3. The component is in or attached to a Risk Category IV structure, and it is needed for continued operation of the facility or its failure could impair the continued operation of the facility.
4. The component conveys, supports, or otherwise contains hazardous substances and is attached to a structure or portion thereof classified by the Authority Having Jurisdiction as a hazardous occupancy.

All other components shall be assigned a component Importance Factor, I_p , equal to 1.0.

13.1.4 Exemptions. The following nonstructural components are exempt from the requirements of this chapter:

1. Furniture except storage cabinets, as noted in Table 13.5-1;
2. Temporary or movable equipment;
3. Architectural components in Seismic Design Category B, other than parapets, provided that the component Importance Factor, I_p , is equal to 1.0;
4. Mechanical and electrical components in Seismic Design Category B;
5. Mechanical and electrical components in Seismic Design Category C provided that either

- a. The component Importance Factor, I_p , is equal to 1.0 and the component is positively attached to the structure; or
 - b. The component weighs 20 lb (89 N) or less or, in the case of a distributed system, 5 lb/ft (73 N/m) or less.
6. Discrete mechanical and electrical components in Seismic Design Categories D, E, or F that are positively attached to the structure, provided that either
- a. The component weighs 400 lb (1,779 N) or less, the center of mass is located 4 ft (1.22 m) or less above the adjacent floor level, flexible connections are provided between the component and associated ductwork, piping, and conduit, and the component Importance Factor, I_p , is equal to 1.0; or
 - b. The component weighs 20 lb (89 N) or less or, in the case of a distributed system, 5 lb/ft (73 N/m) or less; and.
7. Distribution systems in Seismic Design Categories D, E, or F included in the exceptions for conduit, cable tray, and raceways in Section 13.6.5, duct systems in 13.6.6 and piping and tubing systems in 13.6.7.3. Where in-line components, such as valves, in-line suspended pumps, and mixing boxes require independent support, they shall be addressed as discrete components and shall be braced considering the tributary contribution of the attached distribution system.

13.1.5 Premanufactured Modular Mechanical and Electrical Systems. Premanufactured mechanical and electrical modules 6 ft (1.8 m) high and taller that are not otherwise prequalified in accordance with Chapter 13 and that contain or support mechanical and electrical components shall be designed in accordance with the provisions for nonbuilding structures similar to buildings in Chapter 15. Nonstructural components contained or supported within modular systems shall be designed in accordance with Chapter 13.

13.1.6 Application of Nonstructural Component Requirements to Nonbuilding Structures. Nonbuilding structures (including storage racks and tanks) that are supported by other structures shall be designed in accordance with Chapter 15. Where Section 15.3 requires that seismic forces be determined in accordance with Chapter 13 and values for R_p are not provided in Table 13.5-1 or 13.6-1, R_p shall be taken as equal to the value of R listed in Chapter 15. The value of a_p shall be determined in accordance with footnote *a* of Table 13.5-1 or 13.6-1.

13.1.7 Reference Documents. Where a reference document provides a basis for the earthquake-resistant design of a particular type of nonstructural component, that document is permitted to be used, subject to the approval of the Authority Having Jurisdiction and the following conditions: